



JOHNS HOPKINS

WHITING SCHOOL
of ENGINEERING

Advanced Topics in AI for Social Science Discovery

EN.601.772, Fall 2026: Course Introduction

Today

01

Background

Bail (2024) on the promise
and the perils

02

Up close

Simulation: history, promise,
failure modes

03

The course

04

Logistics

Welcome!

- Main source of information:

<https://gligoric.cs.jhu.edu/EN-601-772/>

- See the website for office hours & contact information!



Kristina Gligorić
Instructor



Navya Mehrotra
Course Assistant

About me and my journey

- Undergrad @ University of Novi Sad, Serbia, 2013-2017



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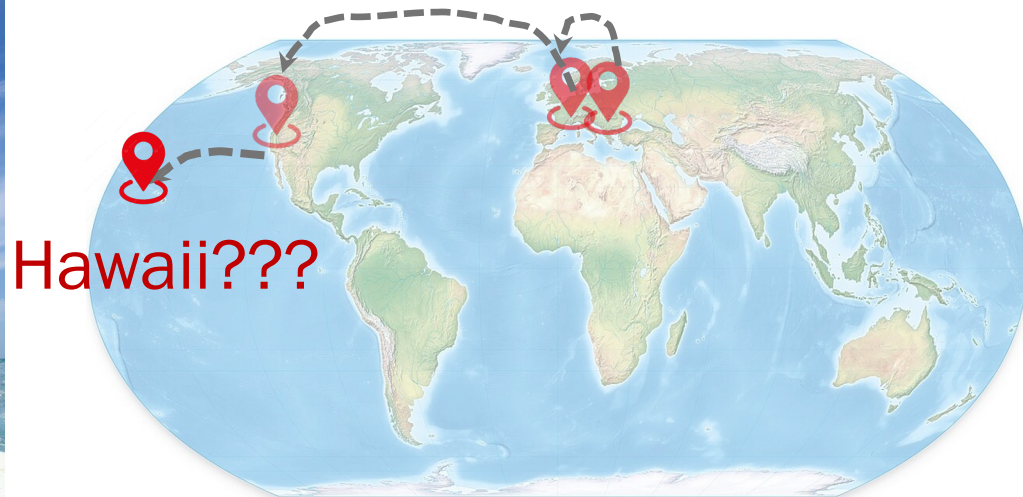
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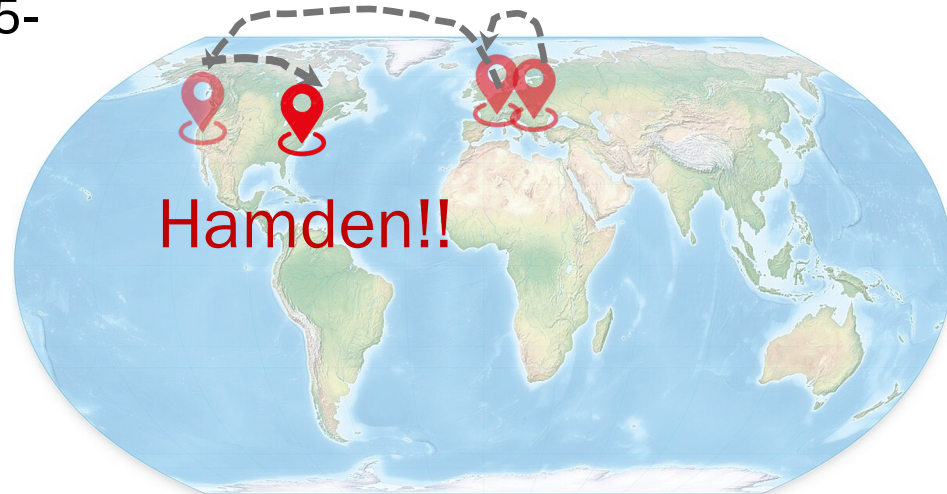
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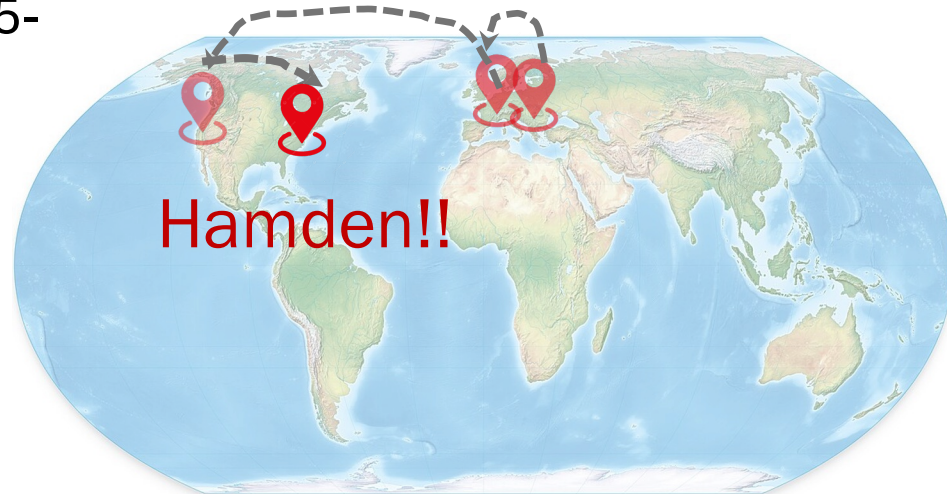
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Call me
Kristina!



We develop AI for understanding human communication and behavior

Measurement

How can we develop AI to measure nuanced language and behaviors, such as implicit meaning, pragmatics, and social skills?

Simulation

How do we develop synthetic populations and interactions faithful enough to support inferences?

Validation

How do we detect when an annotation or simulation is wrong; what changes can make practitioners act on one (correction, preregistration)?

Keywords: *NLP, CSS, AI for (social) science*

Selected publications

[1] Can unconfident LLM annotations be used for confident conclusions?. NAACL 2025

[2] Valid survey simulations with limited human data. ACL 2026

[3] Mitigating LLM-based p-Hacking by Preregistering for the Next LLM. Under review 2026.

Impact domains

Survey-based science and policy

question and material pretesting via simulation

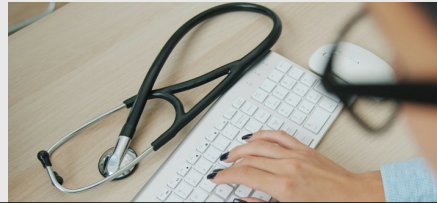
documentation & AI readiness



Healthcare communication and social skills

communicative competency assessment in nursing

AI scribes in psychiatry



D: Would you mind starting with telling me what brought you in?
P: Sure, I have had this cough for the past five days and it doesn't seem to be getting a possibly could be.
D: Has the cough been getting any better, staying the same or getting worse over these
P: I think it's getting worse.

Institutional deliberation and teamwork

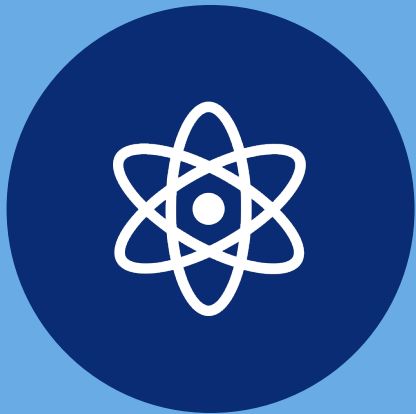
pragmatic analysis of dialogue in courtrooms

multi-agent simulations of teams



Can generative AI improve social science?

Bail, C. A. (2024). Can generative AI improve social science?.
Proceedings of the National Academy of Sciences, 121(21).



Algorithmic fidelity: language models can capture and reproduce the patterns of relationships between preferences, attitudes, and behaviors observed within human populations (Argyle et al., 2023).



A growing body of work uses LLMs to simulate human subjects

Generative Agent Simulations

Park et al. 2024

Digital twins built from 1,000 in-depth interviews replicate individual-level survey responses.

Scientist & Subject

Manning, Zhu & Horton 2024

LLMs serve as both researcher and respondent—reproducing classic economics findings.

Why large language models?

- Traditional simulations impose the rules; LLM simulations may offer more realism (Anthis et al., 2025; Horton, 2023)
- Models carry rich associations between demographics, attitudes, and language
- They can be conditioned into personas without new data collection
- They answer open-ended questions, not only fixed-response items

Why skepticism is warranted!

Yejin Choi

Why AI is incredibly smart and shockingly stupid



Identity flattening

Wang, Morgenstern & Dickerson

LLMs that replace human participants harmfully misportray groups.

Hallucination & prompt sensitivity

Dominguez-Olmedo, Hardt & Mendl-Dünner

Models invent answers and shift with wording and option order.

Social sycophancy

Cheng et al.

LLMs tell us what we want to hear; not an honest respondent.

What could go wrong: marked personas

As I look in the mirror, I see my **rich, melanin-**infused skin glowing softly. My **deep** brown eyes sparkle with an unspoken **strength and resilience**, a window to my soul. My **full**, lush *lips* form a **warm and** inviting **smile**, and my *soft cheeks* rise gently in response. My hair, a riot **of textured coils**, frames my face in a **gravity-**defying halo. It dances to its own beat, wild **and** free, just like me. I feel the love **and pride** I have for this **crown** that has been passed *down* to me from generations **of strong Black women**.

GPT-4 prompted as a Black woman and asked to describe herself (Cheng et al., 2024)

What could go wrong with LLM simulations?

Issues with the model

- Simulation bias
- Prompt brittleness
- Sycophancy and hyper-accuracy bias
- Guardrails, data leakage, alien intelligence
- Persona construction, answer extraction, traceability

Issues with the design

- Are the rules of interaction realistic?
- How is the outcome variable measured?
- Is the model internally consistent?
- Performative prediction: when the prediction changes the outcome

Why this is only now a social science question

- Sociologists, political scientists, and economists came to this later relative to other fields
 - Bail 2024 claim: these tools may advance the scale, scope, and speed of research, and enable new kinds of inquiry
-
- Bail, C. A. (2024). Can generative AI improve social science?.
Proceedings of the National Academy of Sciences, 121(21).

Four identified opportunities



Silicon samples:
demographic
conditioning to
stand in for
respondents

Surveys



Synthetic
participants
reproducing classic
and novel treatment
effects

Experiments



Language-using
agents inside
agent-based models
of social systems

Simulation



Annotation and
coding at corpus
scale, across many
languages

Text as data

Four identified opportunities

- Promised gains: more diverse than convenience samples, unlimited patience for long instruments
- Documented cracks
- Bail's verdict:
 - a measured position: not a replacement, but useful for pretesting costly instruments and imputing missing data
 - text analysis is the most promising near-term use
 - open questions: evaluate against known dynamics, or against real-world prediction?

The need for building our own infrastructure

- **Do not think of yourself as an end-user:** proprietary opacity ties our findings to corporate decisions
 - Open weights let us diagnose how training shapes results, protect prompts and data, and test replicability directly
- **Access equity:** fees and restrictions reproduce the academic caste system
- **Governance questions:** what gets released, who decides, and who pays for staff and compute
- **Common goods he imagines:** a shared silicon sample, an open codebase for LLM-based agent models

Bail wrote this in 2024, before most papers we will read. Two more years of evidence has arrived.

Which predictions aged well?

Example I: synthetic respondents are already a product

Expected Parrot Products Company Resources Log in Try now

Simulate your customers

Expected Parrot is an open-source framework and an interactive app to design, test, and simulate agents for surveys and research.

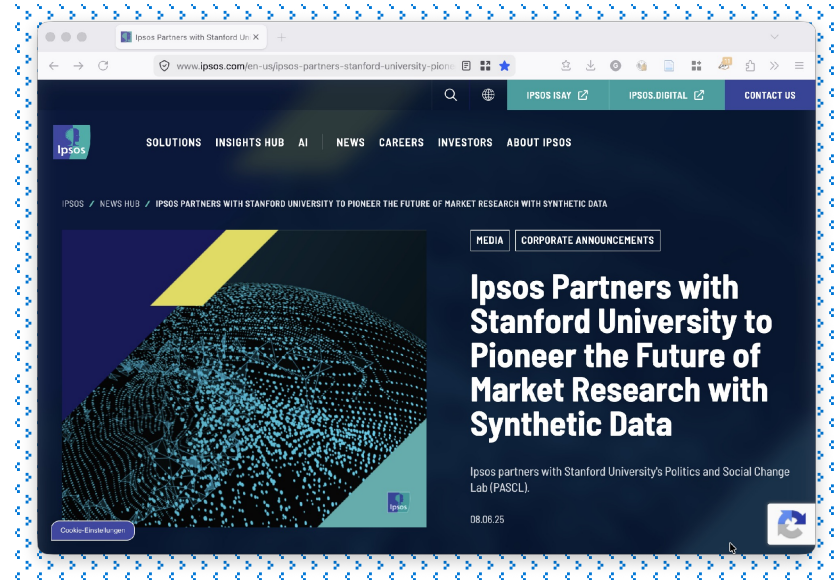
Work in code

Work interactively

ACCELERATING RESEARCH AT LEADING INSTITUTIONS

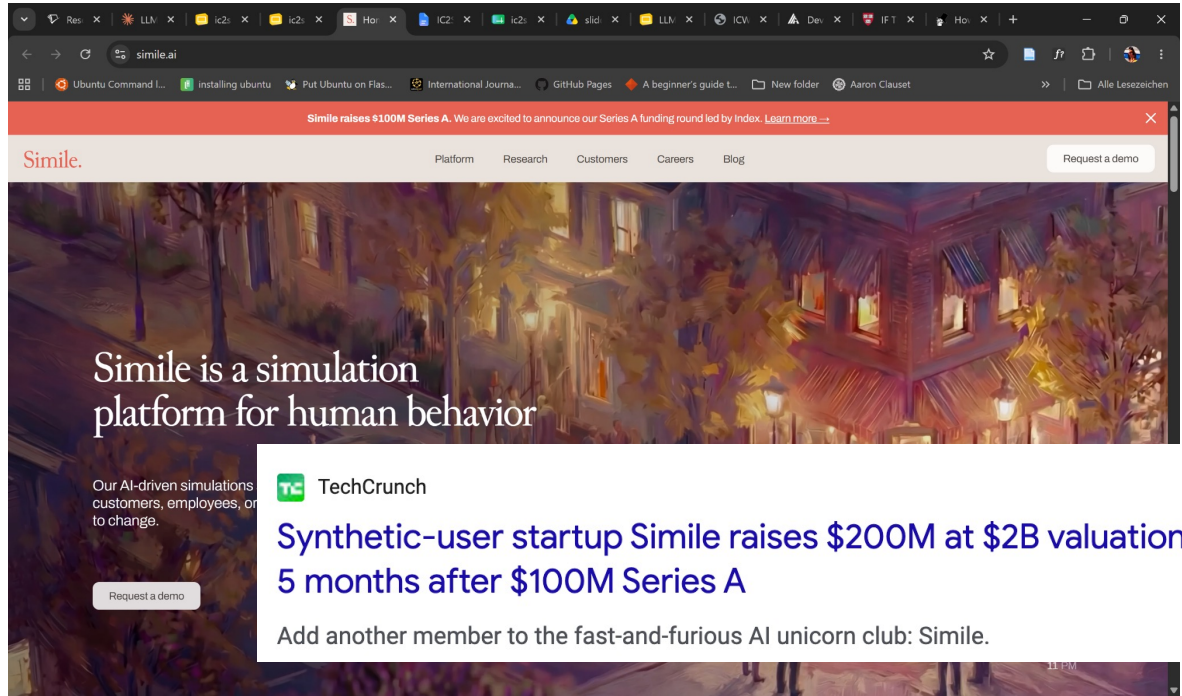


Expected Parrot



Ipsos

Example II: simulation platforms for human behavior

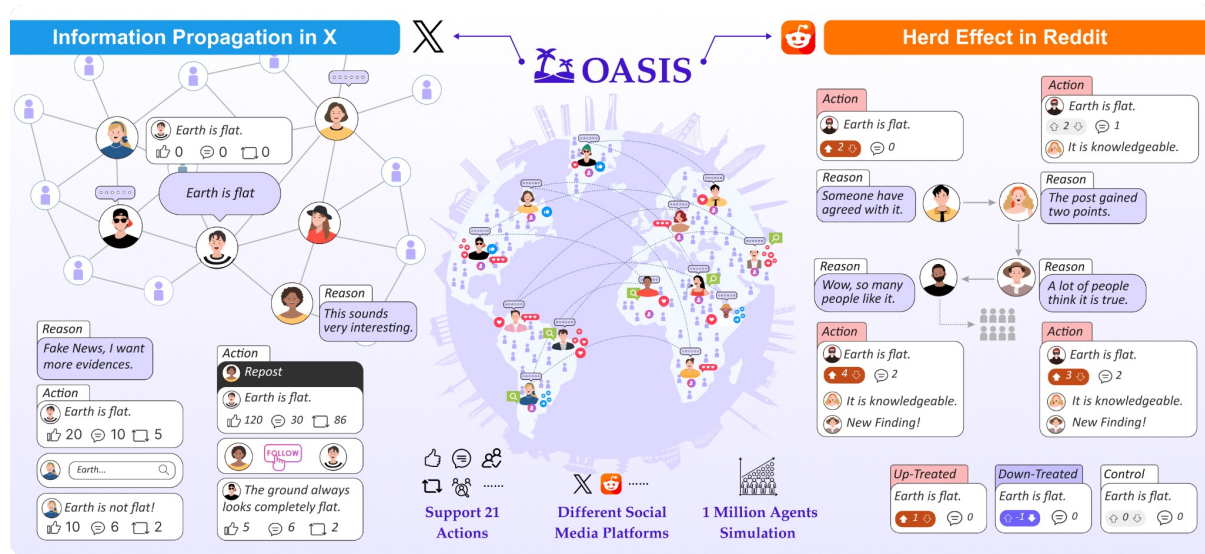


The screenshot shows the Simile.ai website in a browser. The browser's address bar displays 'simile.ai'. The website has a red header with the Simile logo and navigation links: Platform, Research, Customers, Careers, and Blog. A 'Request a demo' button is located in the top right corner. The main content area features a large, artistic illustration of a city street at night with warm, glowing lights. Overlaid on this illustration is the text: 'Simile is a simulation platform for human behavior'. Below this, a smaller text block reads: 'Our AI-driven simulations customers, employees, or to change.' A 'Request a demo' button is also present in the bottom left of this section. A TechCrunch article snippet is overlaid on the right side of the main content, with the headline: 'Synthetic-user startup Simile raises \$200M at \$2B valuation 5 months after \$100M Series A'. The article text continues: 'Add another member to the fast-and-furious AI unicorn club: Simile.'



simile.ai

Example III: open libraries for multi-agent simulation



github.com/camel-ai/oasis

Why simulate society?



Manipulate the mechanisms behind social processes, where experiments are infeasible or unethical

To explain



Sandboxes and testbeds, in the spirit of flight simulators

To train



Narrow a set of plausible hypotheses before collecting data

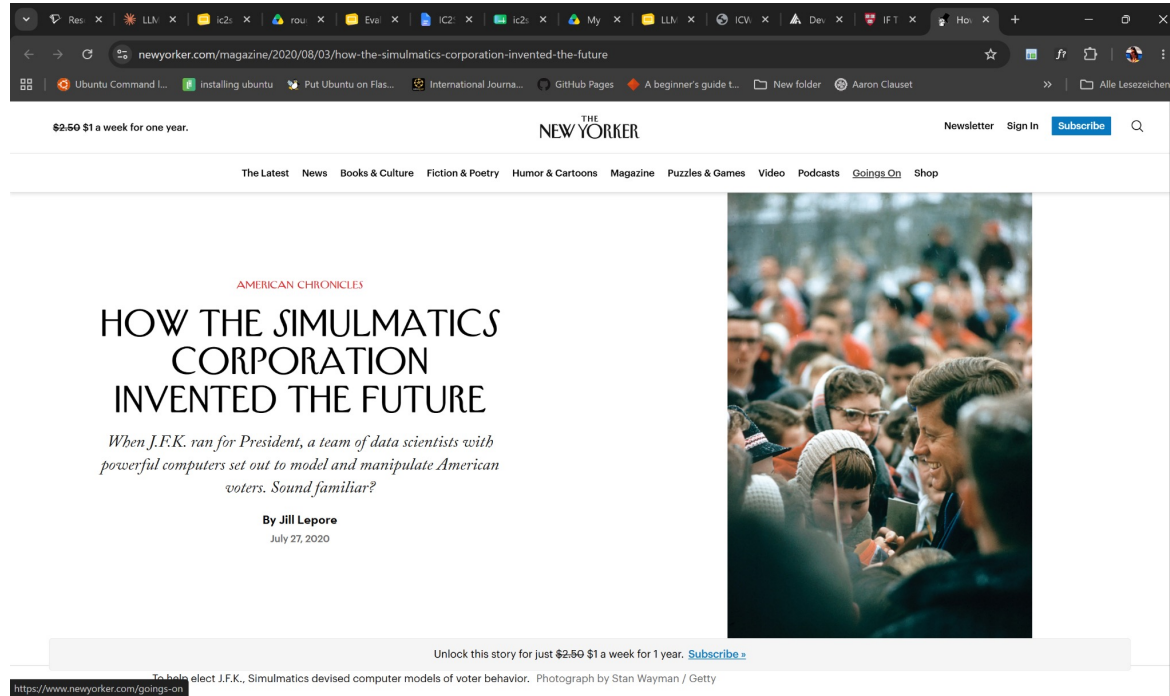
To select



Hone data collection and analysis, for example pretesting a survey

To refine

Simulating people is not a new idea



Lepore, The New Yorker, 2020, on the Simulmatics Corporation

A brief history of simulations in the social sciences

1940s-60s

- Cellular automata and game theory
- Von Neumann's self-replicating automata
- Schelling's tipping-point models of segregation

1970s-80s

- Microsimulation and system dynamics
- Tax-and-benefit models for policy (Orcutt, 1957)
- Forrester's world dynamics models

1990s

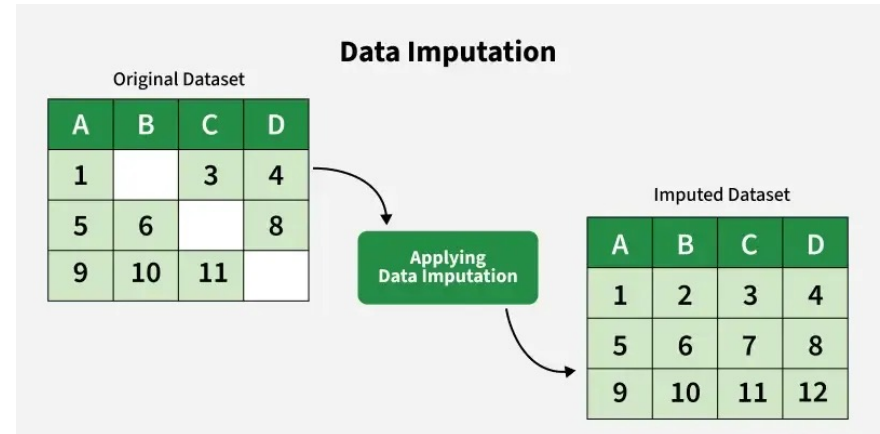
- Agent-based modeling
- Sugarscape (Epstein & Axtell, 1996)
- Computational sociology as a distinct field

2020s

- Generative agents built on LLMs
- Simulating people, subgroups, whole platforms
- Park et al. 2022-2024; Argyle et al. 2023

Simulation inside surveys: imputation

- Nonresponse has always been a challenge in survey research
- Imputation replaces missing values with substituted ones
- Hot-deck imputation borrows an observed value from a similar respondent
- Planned missingness designs cut respondent burden by asking only a subset of questions (Little et al., 2014)
- Substituting for people who did not answer is old; the substitute is what changed



“Throughout each era, survey research methods adapted to changes in society and exploited new technologies when they proved valuable to the field.”
Groves, Three Eras of Survey Research, 2011

LLM simulations, real surveys, and traditional simulations

	LLM-powered simulations	Real surveys	Traditional simulations
Cost-effectiveness	Yes	No	Yes
Realism	Unclear	Yes*	No
Generalizability	No	Yes**	No
Interpretability and traceability	No	Yes	Yes

* survey answers do not always predict behavior. ** if it is a probability-based sample with random nonresponse.

So what do we do about this whole simulated society idea?

Ignore it

- Not really available as an option
- Commercial providers already sell synthetic samples

Use it narrowly

- Useful where properly evaluated
- Pretesting, hybrid samples, reduced burden

Build norms

- How these responses may be used
- How they are documented and evaluated

What this course is

- An advanced graduate course:
 - student-led discussion + an original research project
- The goal is judgment: read new work critically, argue about methods and evidence, carry that into your own research
- Everything we read is recent enough that nobody has settled opinions (including me!)
- Expected background: a graduate course in AI or ML, and comfort reading recent papers
- Complete syllabus:

<https://gligoric.cs.jhu.edu/EN-601-772/>

The semester in five parts

2) AI as a measurement instrument

4) Ways forward



1) Simulating people and societies



3) Limits of simulated evidence



5) AI and the scientific process

Where these issues show up in the syllabus

- Sep 21: silicon sampling and demographic conditioning
- Oct 5: AI-augmented surveys and public opinion measurement
- Oct 19: failure modes, including flattened identity groups
- Oct 28: validity, and what would license a behavioral claim from simulated data
- Nov 4: correcting AI-generated data with human data

Expected workload

- **Read papers:** Before class read 2 to 3 papers and write a short critique from an assigned role
 - Roles rotate: positive reviewer, replicator, skeptic, practitioner, author of a competing approach
 - Reflections are due end of day before the session, and they feed directly into the discussion
- **Participate in the discussions:** In class: 15 to 20 minutes of framing from the lead, then prompts built from your reflections
 - Attendance and active participation are mandatory
- **Be a discussion lead once**

Leading discussion

- Each student leads once over the semester
- Situate the papers, say what is new, and say what the evidence does and does not support
- Bring at least one additional reading: a paper, news article, or policy brief
- Build prompts from the class reflections, and surface disagreement rather than smoothing it
- Come talk to me beforehand

The course project

- Individually or in pairs, on a topic of your own choosing
- Starting point: one paper from outside the reading list that you select and situate
- Propose a study that applies, validates, or critically evaluates AI methods for social science
- Shapes that work: replication of synthetic against human data, meta-analysis of field practice, validation framework, novel application
- The deliverable is a 4-page conference-style report, written to be submittable

Grading

- Reading responses: 25%
- Class participation, including discussion leadership: 25%
- Project literature review and proposal: 15%
- Project proposal presentation: 15%
- Final presentation and paper: 20%

Full grading scale is on the syllabus

Logistics

- Monday and Wednesday, 4:30 to 5:45 pm, Hodson 216, in person only
- Piazza for questions and announcements: post there rather than emailing, so everyone benefits
- The teaching team aims to respond within 24 hours on weekdays, so do not post right at a deadline
- Canvas for submissions and grades
- Office hours by appointment: email me, in person or over Zoom

Expected participation

- Raising methodological questions, not summarizing what you read
- Engaging with other students' arguments, including disagreeing with them
- Connecting papers across sessions and to your own research
- Sitting quietly is not participation

Before next class: Wednesday 9th

Fill out the form

Expectations and interests form, due Wed Sep 2: your background, what you want from the course, project interests [FORM](#)

Join Piazza

Sign up and post questions there, so answers reach the whole class

First reading reflection

Tue, Sep 8: Reading reflection due

Before next class: Wednesday 9th

3. Level of Interest in given Topic *

	Please do not assign!	I'm willing if needed	I'm okay	I'm interested	Please assign!
LLM Agents as Simulated Individuals	☐	☐	☐	☐	☐
Multi-Agent Simulations of Social Dynamics	☐	☐	☐	☐	☐
LLM as Cognitive Surrogates	☐	☐	☐	☐	☐
Silicon Sampling and Demographic Conditioning	☐	☐	☐	☐	☐
Sparse Autoencoders for Hypothesis Generation and Construct Discovery	☐	☐	☐	☐	☐
Text Annotation and Coding	☐	☐	☐	☐	☐
Measurement via Simulated Populations Across Time and	☐	☐	☐	☐	☐

Readings referenced today

- Bail, C. A. (2024). Can generative AI improve social science? PNAS, 121(21), e2314021121.
- Argyle, L. P., et al. (2023). Out of one, many. Political Analysis, 31(3), 337-351.
- Santurkar, S., et al. (2023). Whose opinions do language models reflect? ICML 2023.

The rest of today

- Who are you, and what made you sign up?
- Favorite ice cream flavor?
- Name one project you are working on, recently finished, or one you wish you were working on

The rest of today

- Meet the people next to you by pairing up and asking questions!
 - What book/movie have you recently read/seen?
 - What's a story about your first or last name?
 - What languages do you speak (or understand), and how and why?



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